R101/A | Two-stage provenance

Fixed TRAIN hint endpoint, then independently reset original full18 learning.

Teacher: random. Stage I:12800 updates, logical batch1024, seed2; CVRP+15MVRP TRAIN81000/task only.

Separate non-affine per-node LayerNorm MSE; one dropout forward. No outcome labels, learned adapter or teacher VAL/TEST. Fixed endpoint, not validation selected.

Stage II: fresh original R84/R88B student; ONLY instance_encoder transferred. All non-encoder parameters remain identical fresh R84. Optimizer, scaler, RNG and task/sample streams reset.

Original1842953 parameters; no added heads or features. R77 CE .35 / comparison .35 / risk .02 / R-Drop .35. No hint loss, teacher arrays or online donor in Stage II or deployment.

Both A/B have12800+57600 updates; R89B/R84A are NOT total-budget-matched. B teacher has additional external solver pretraining history. The paper memo provides conditional motivation only.

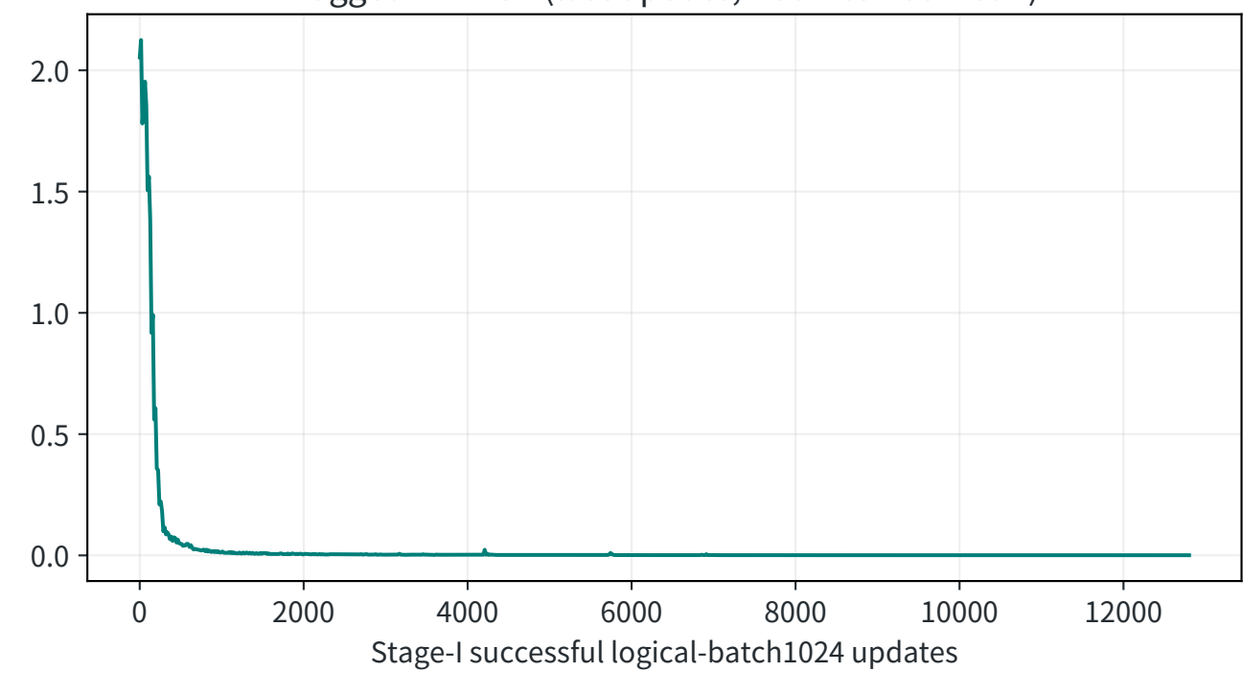
Native/FP64 ind winners; legal-pool Oracle and TRAIN-fitted SBS; ALL is task-equal. Development-exposed TEST; no TEST/LIB checkpoint selection.

Stage-I encoder SHA256: 1267da285be94fe4ff642cffcf4bac3b84d5bfb2ce6f6b5b255af3308dab02bb

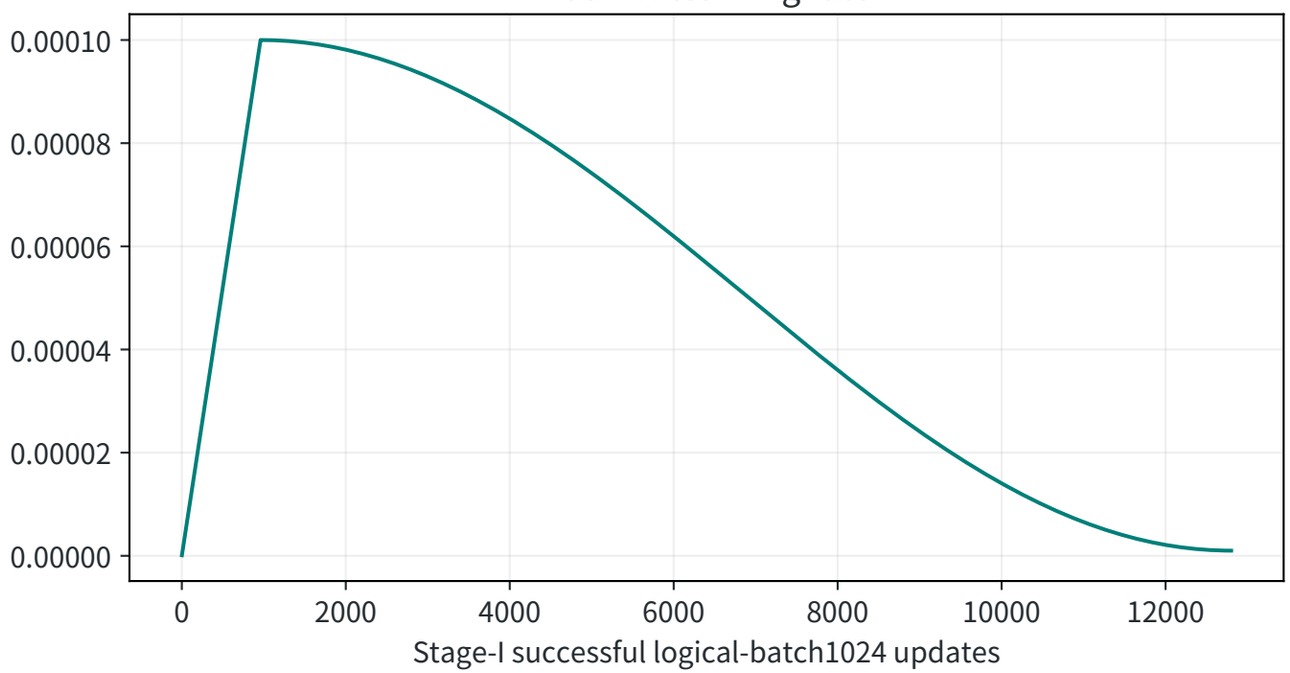
Unchanged non-encoder SHA256: 970a8aa2a548f9034f34b73d060662f71808e46359335185ce07bfc5b1bb2fa6

R101/A | TRAIN-only random node hints; fixed12800 endpoint

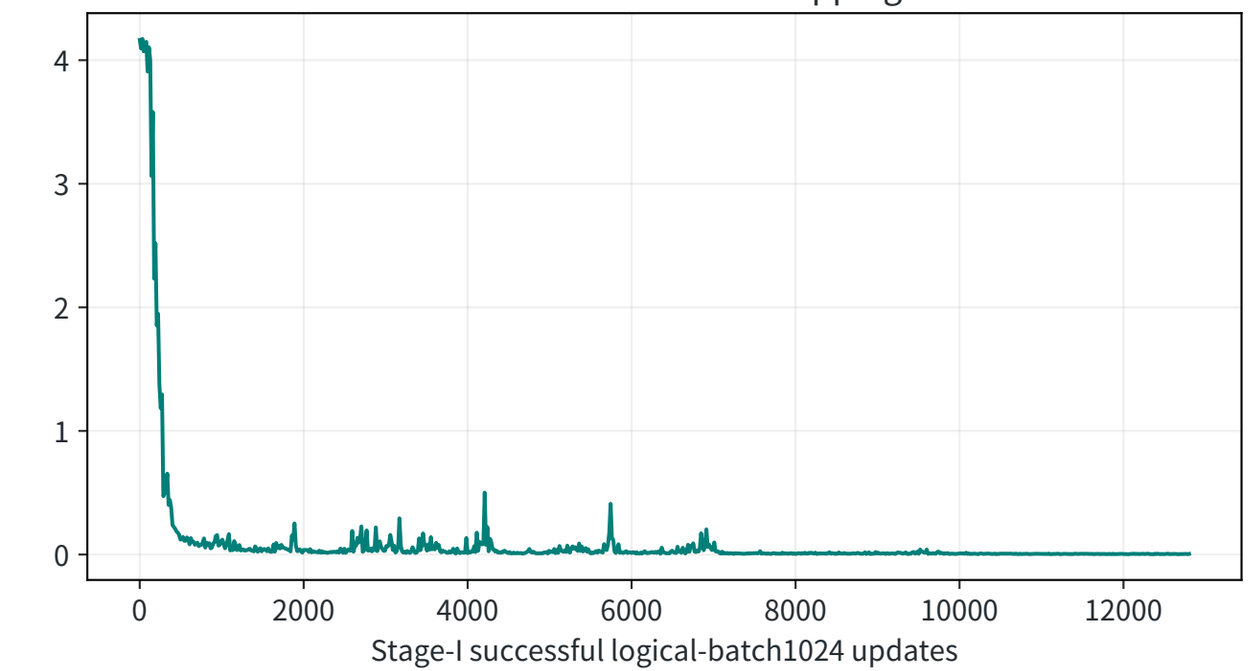
Logged LN-MSE (last update, not interval mean)



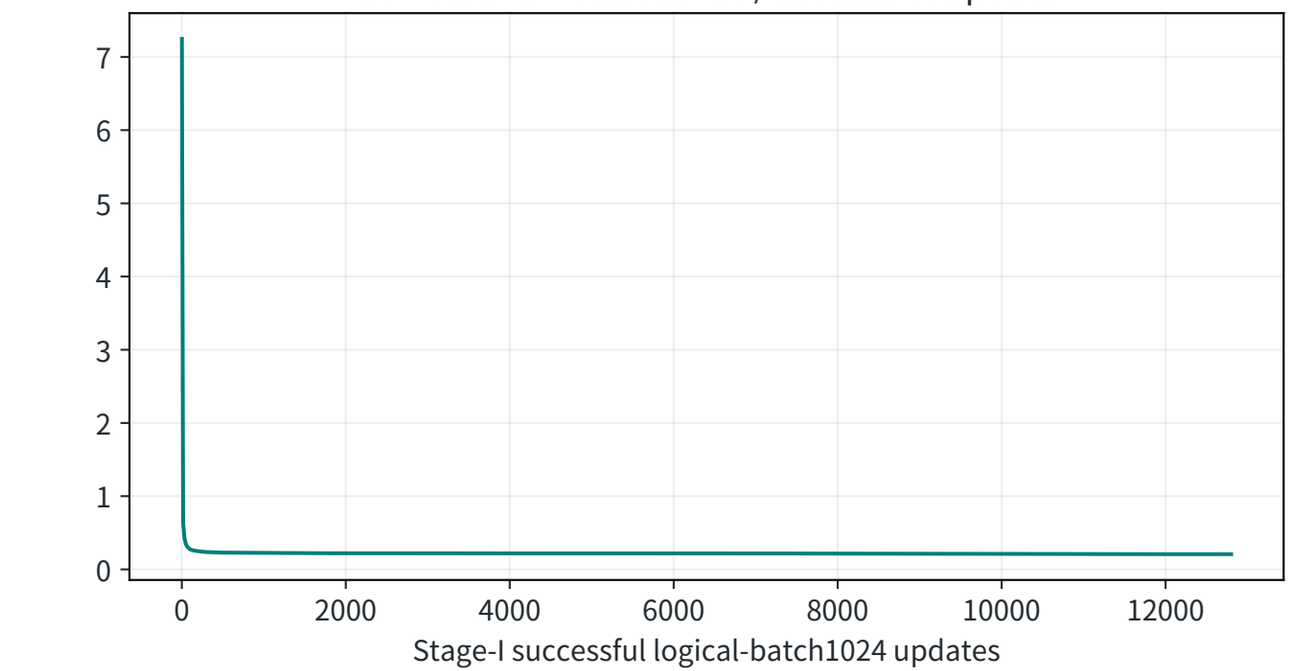
AdamW learning rate



Gradient norm before clipping



Cumulative wall seconds / successful update



R101/A | Locked full18 results

Same unique VAL checkpoint on VAL/TEST/LIB; native/FP64 ind; historical NSS weights disclosed.

epoch 29 | update 41760 | seed2 | fresh initialization

Data: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
VAL ALL	71.1519	66.2037	11.416572	11.427978	11.371692	-0.1949	+0.3927	0.4026
TEST ALL	71.3222	66.1802	11.373839	11.385299	11.329932	-0.1979	+0.3853	0.3930

Dataset	n	NSS Top1 (%)	R101 Top1 (%)	NSS cost	R101 cost
TSP TEST	4500	45.888889	47.000000	7.970400719	7.970917341
CVRP TEST	4500	45.977778	48.200000	18.004802368	18.009998522
TSPLIB	49	20.408163	28.571429	8.217907074	8.210717059
CVRPLIB	100	70.000000	62.000000	67.101111717	67.254707232

NSS: R79 weights trained on historical70114 TRAIN, selected using historical VAL; not retrained on new90k.

NSS TEST predictions were freshly computed on these4500 instances; old3895 TEST metrics are not reused.

SBS is fitted on the new81000 TRAIN only. Oracle is the per-instance legal-pool minimum.

ALL is the equal-problem mean; mixed raw cost is descriptive. TEST informed prior research.

Method selection and true winner use separate colors on the same axes; Top1 is per-instance accuracy.

Selection head: original_score_mlp_maximin | objective: original R77 native CE

R101/A | VAL: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.1519	66.2037	11.416572	11.427978	11.371692	-0.1949	+0.3927	0.4026	per-problem TRAIN SBS
TSP	46.7333	33.1111	8.024500	8.045354	7.990533	-0.2592	+0.4251	0.4144	T2T
CVRP	46.7556	18.7111	18.159052	18.269377	18.064087	-0.6039	+0.5257	0.7022	RELD_CVRP
ATSP	77.2222	47.4889	1.638651	1.676905	1.632471	-2.2812	+0.3786	0.3745	UNICO_MatPOENet
OVRP	41.1556	35.7556	7.462603	7.469579	7.415777	-0.0934	+0.6314	0.6447	MoSES_RF
VRPB	81.1111	80.8444	9.354106	9.354876	9.318750	-0.0082	+0.3794	0.3873	reld_mt
VRPL	38.2000	33.0889	12.432087	12.437959	12.383444	-0.0472	+0.3928	0.3996	MoSES_RF
VRPTW	79.9333	79.9556	18.819353	18.819431	18.766139	-0.0004	+0.2836	0.2986	reld_mt
OVRPTW	83.8444	83.7778	10.142885	10.143730	10.116859	-0.0083	+0.2573	0.2546	reld_mt
OVRPB	88.5333	88.5111	6.357159	6.357254	6.341078	-0.0015	+0.2536	0.2637	reld_mt
OVRPL	41.7333	35.9111	7.437128	7.445709	7.390416	-0.1153	+0.6321	0.6473	MoSES_RF
VRPBL	78.9111	78.5111	9.368180	9.371523	9.328366	-0.0357	+0.4268	0.4291	reld_mt
VRPBTW	78.8000	78.4222	19.944422	19.950430	19.851084	-0.0301	+0.4702	0.4490	reld_mt
VRPLTW	78.8889	78.7556	18.763599	18.765871	18.707843	-0.0121	+0.2980	0.3145	reld_mt
OVRPBL	89.2444	89.2000	6.369733	6.370093	6.352824	-0.0057	+0.2662	0.2774	reld_mt
OVRPBTW	83.6000	83.4444	10.660230	10.661098	10.623245	-0.0081	+0.3482	0.3204	reld_mt
OVRPLTW	83.5778	83.4889	10.156922	10.157423	10.129818	-0.0049	+0.2676	0.2707	reld_mt
VRPBLTW	79.3111	79.4000	19.797034	19.797044	19.706420	-0.0001	+0.4598	0.4473	reld_mt
OVRPBLTW	83.1778	83.2889	10.610650	10.609955	10.571302	+0.0065	+0.3722	0.3517	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R101/A | TEST: ALL and18 tasks

TRAIN-fitted SBS and unchanged native/FP64 costs; no per-task checkpoint assembly.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)	TRAIN SBS
ALL	71.3222	66.1802	11.373839	11.385299	11.329932	-0.1979	+0.3853	0.3930	per-problem TRAIN SBS
TSP	47.0000	34.5333	7.970917	7.989621	7.937593	-0.2341	+0.4198	0.4108	T2T
CVRP	48.2000	17.9333	18.009999	18.124983	17.918509	-0.6344	+0.5106	0.6687	RELD_CVRP
ATSP	76.9333	47.2444	1.635289	1.674207	1.629441	-2.3246	+0.3589	0.3580	UNICO_MatPOENet
OVRP	39.9778	33.6444	7.447082	7.455750	7.400487	-0.1163	+0.6296	0.6388	MoSES_RF
VRPB	80.6222	80.4000	9.350278	9.351810	9.315298	-0.0164	+0.3755	0.3797	reld_mt
VRPL	37.7111	32.2667	12.349928	12.356138	12.300958	-0.0503	+0.3981	0.4060	MoSES_RF
VRPTW	81.1556	80.8444	18.688759	18.690545	18.637408	-0.0096	+0.2755	0.2865	reld_mt
OVRPTW	83.9333	83.9333	10.092823	10.093063	10.068135	-0.0024	+0.2452	0.2363	reld_mt
OVRPB	88.1556	88.1333	6.370594	6.370826	6.352225	-0.0036	+0.2892	0.2953	reld_mt
OVRPL	41.8889	34.6000	7.462971	7.472542	7.418033	-0.1281	+0.6058	0.6150	MoSES_RF
VRPBL	79.5111	79.4222	9.395304	9.396379	9.355967	-0.0114	+0.4205	0.4177	reld_mt
VRPBTW	78.7333	78.7556	19.667108	19.665725	19.577666	+0.0070	+0.4569	0.4527	reld_mt
VRPLTW	79.4000	79.2444	18.801316	18.803136	18.745143	-0.0097	+0.2997	0.3167	reld_mt
OVRPBL	89.2889	89.2889	6.375710	6.375741	6.359310	-0.0005	+0.2579	0.2693	reld_mt
OVRPBTW	84.9111	84.8222	10.659029	10.659591	10.624748	-0.0053	+0.3226	0.2945	reld_mt
OVRPLTW	82.8889	82.8667	10.085770	10.086112	10.059001	-0.0034	+0.2661	0.2668	reld_mt
VRPBLTW	79.7778	79.7333	19.704833	19.707068	19.613284	-0.0113	+0.4668	0.4486	reld_mt
OVRPBLTW	83.7111	83.5778	10.661385	10.662149	10.625574	-0.0072	+0.3370	0.3132	reld_mt

vs_SBS=100*(mean_cost/SBS_cost-1); vs_Oracle=100*(mean_cost/Oracle_cost-1).

actual regret=mean_i[100*(selected_i-oracle_i)/oracle_i]; not a ratio of means.

TRAIN81000 / VAL4500 / TEST4500 per problem. All18 problems have equal macro weight.

R101/A | LIB and locked provenance

Current R101 factory, same buffers and selected checkpoint; no solver execution or LIB tuning.

Split / Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
TSPLIB	28.5714	10.2041	8.210717	8.240435	8.142167	-0.3606	+0.8419	0.8808
CVRPLIB	62.0000	45.0000	67.254707	67.245364	66.945093	+0.0139	+0.4625	0.5744

SHA256: e96d42b96ec03155d8ae60e698459605bcf382e0d9f5d488a224fd809d0e23d9

Unique selection: strict minimum full VAL macro_vs_sbs_pct with common new81000 TRAIN-fitted SBS; first wins ties

Inference: eval mode, original R88B physical TSP node0 slot3; original R77 hand34/methodID, joint2, score MLP/pair differences/maximin; stable ascending global-solver-ID ties; original inputs only; no teacher, hint, adapter or calibration

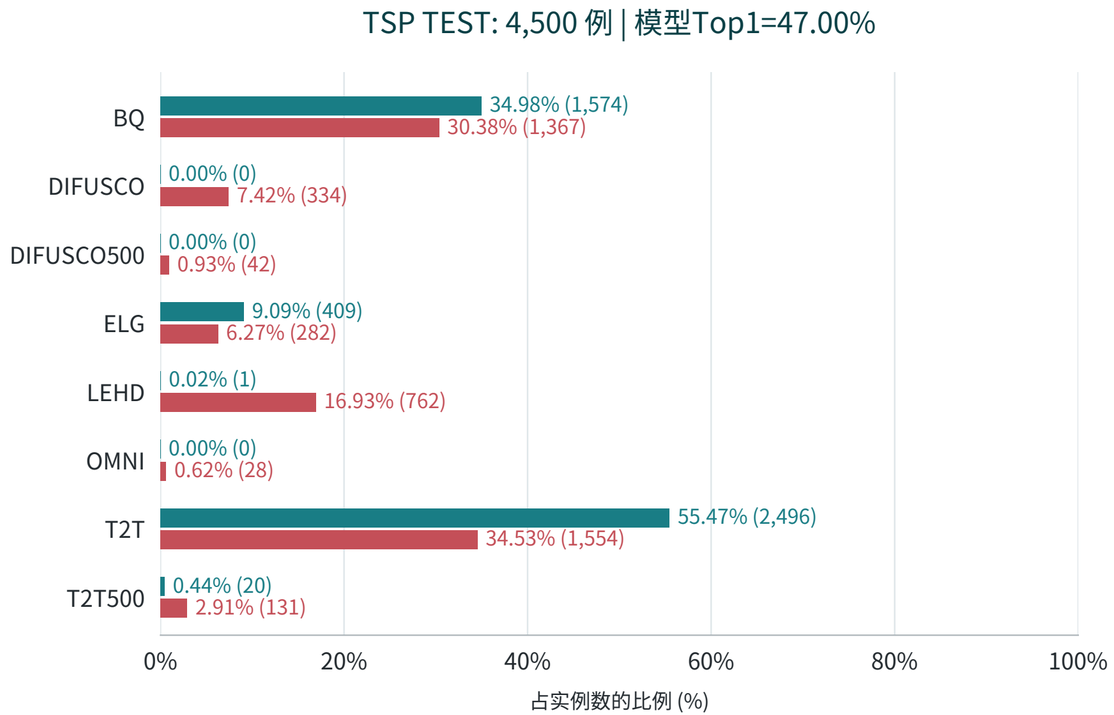
Normalization: new81000 TRAIN only; no TEST/LIB refit.

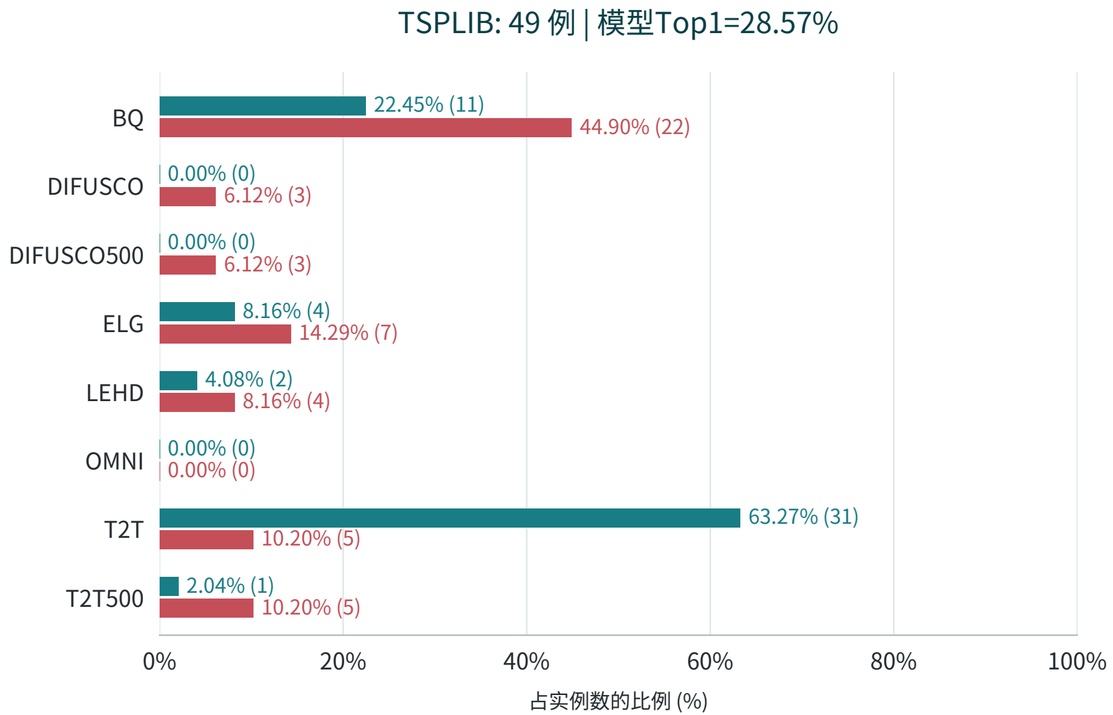
R101/A | TSP | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/TSPLIB | native/FP64 ind





标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

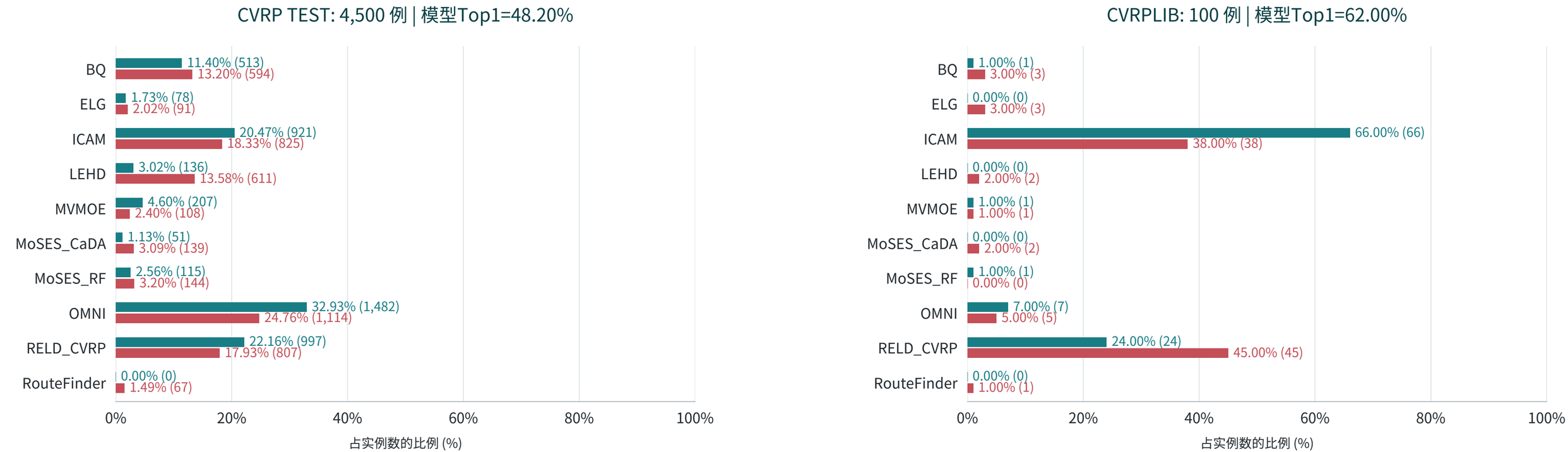
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | CVRP | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

LIB: data/CVRPLIB | native/FP64 ind



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

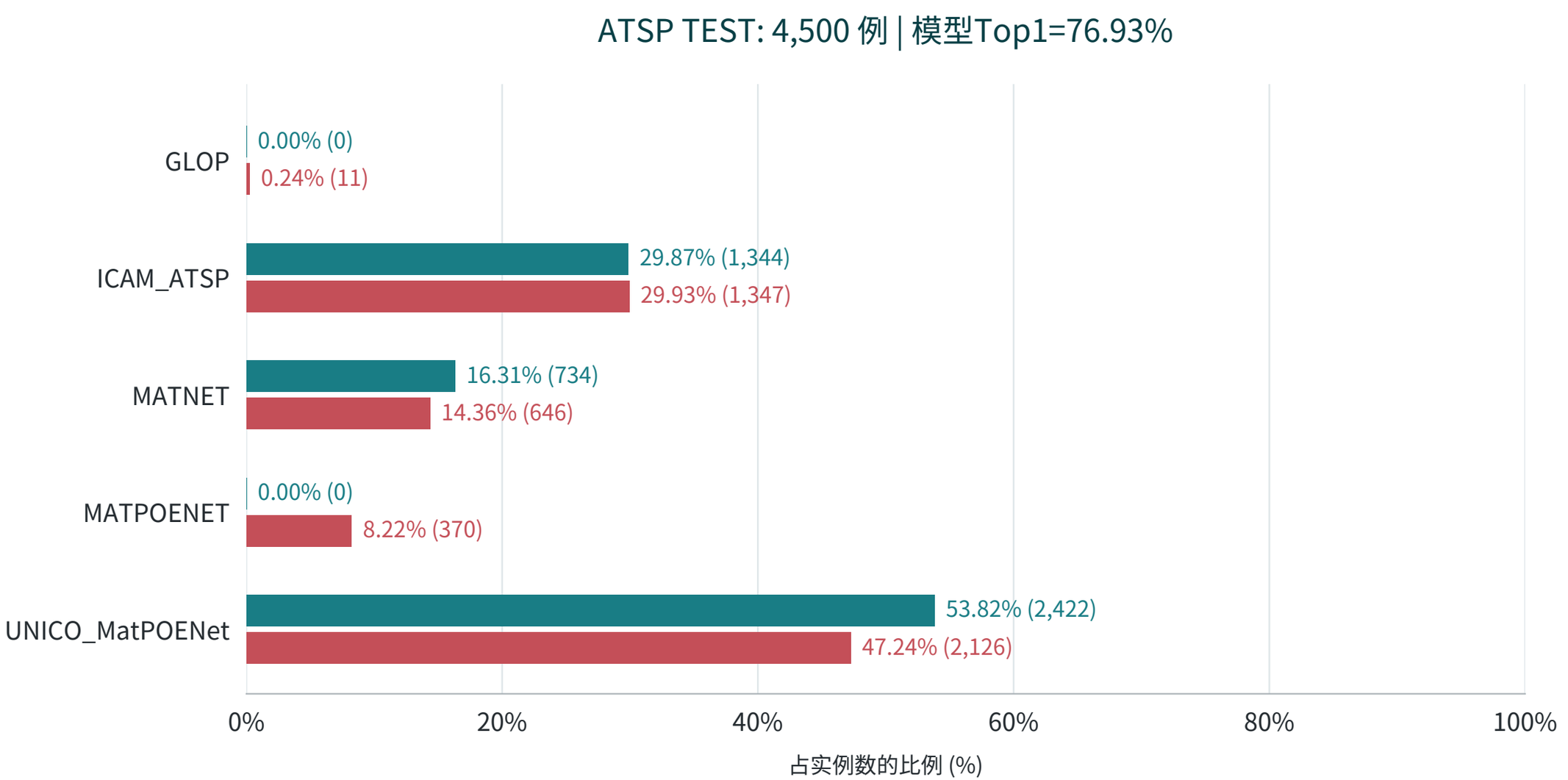
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | ATSP | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

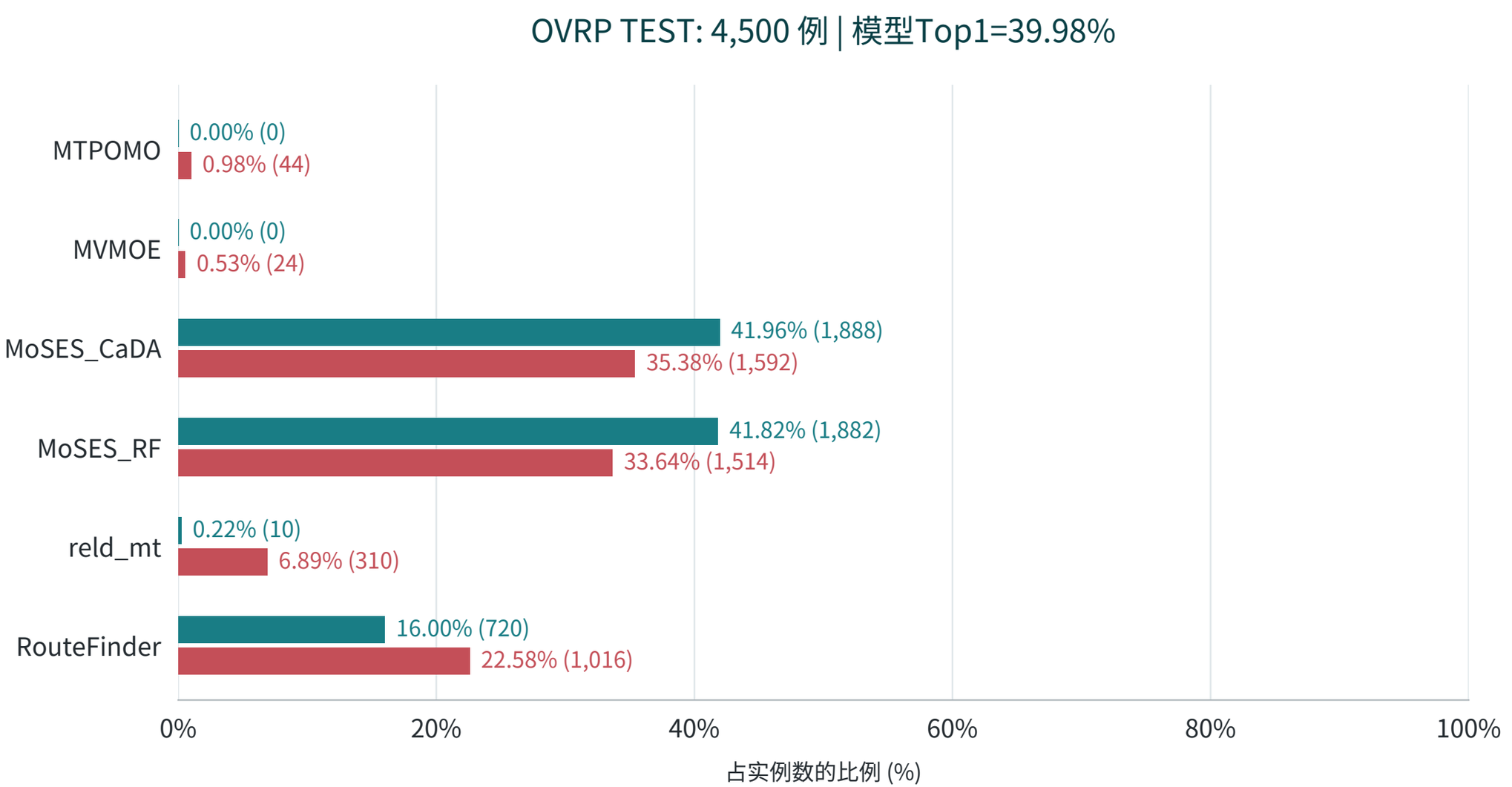
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRP | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

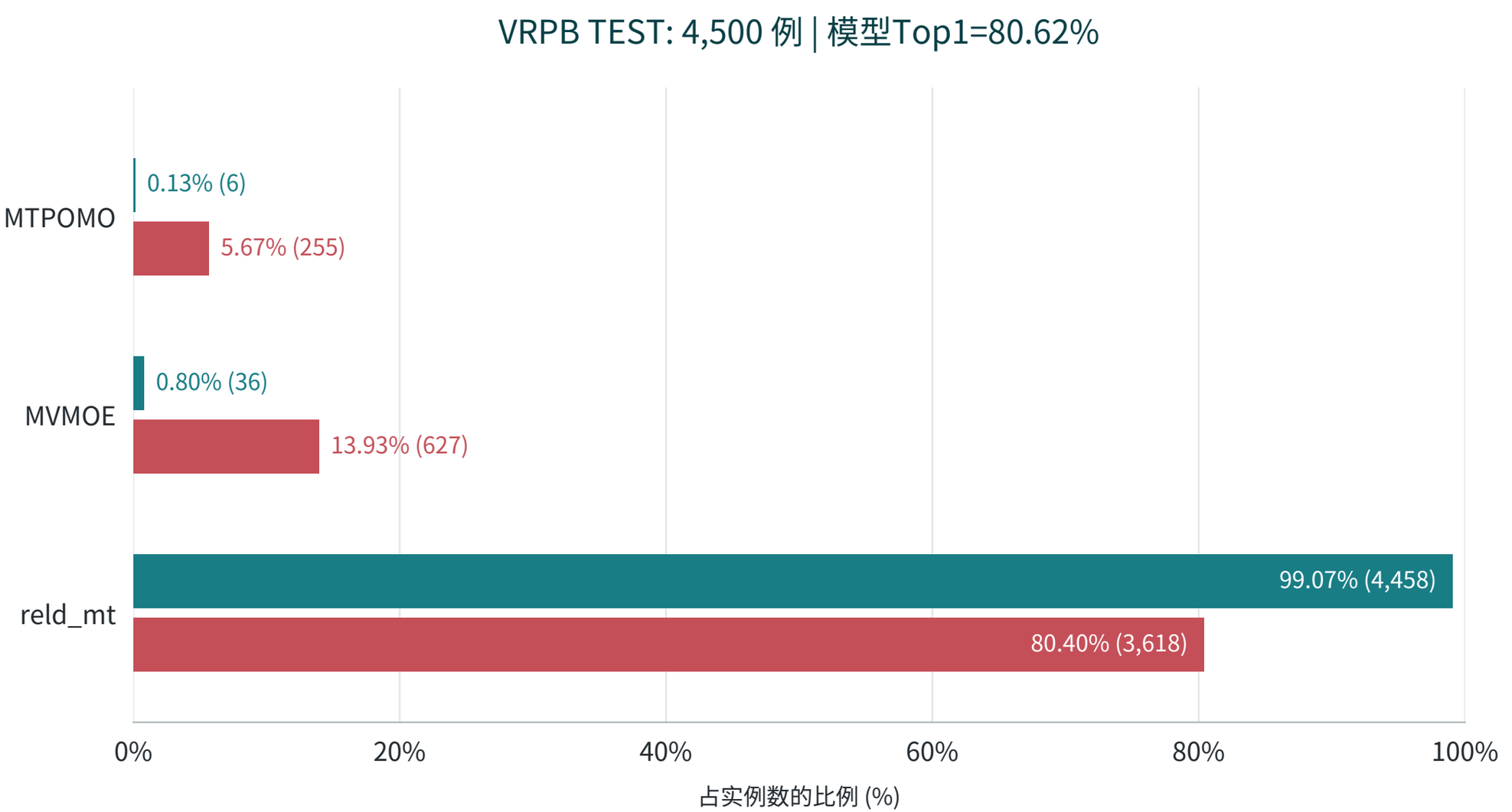
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPB | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

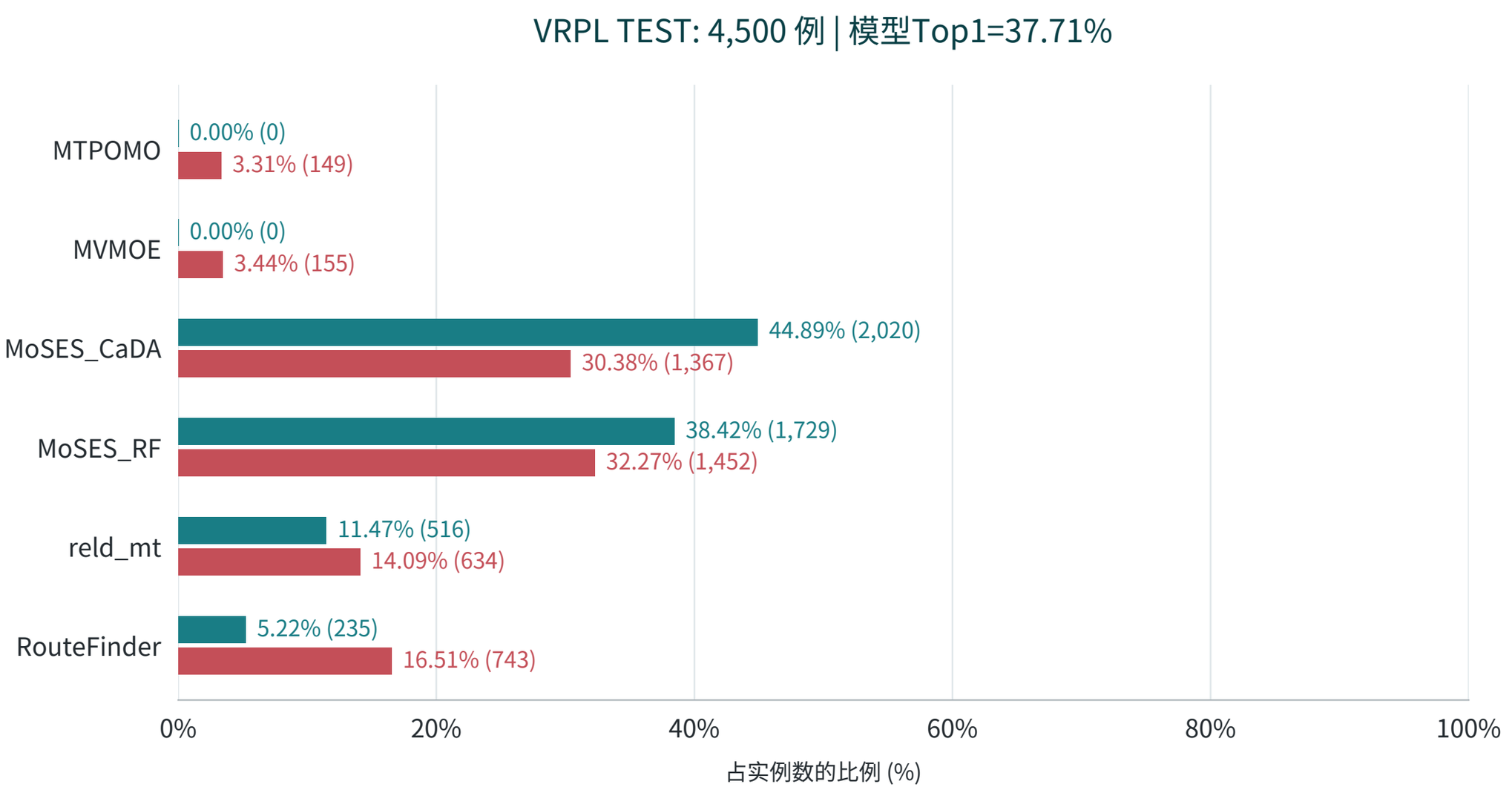
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPL | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

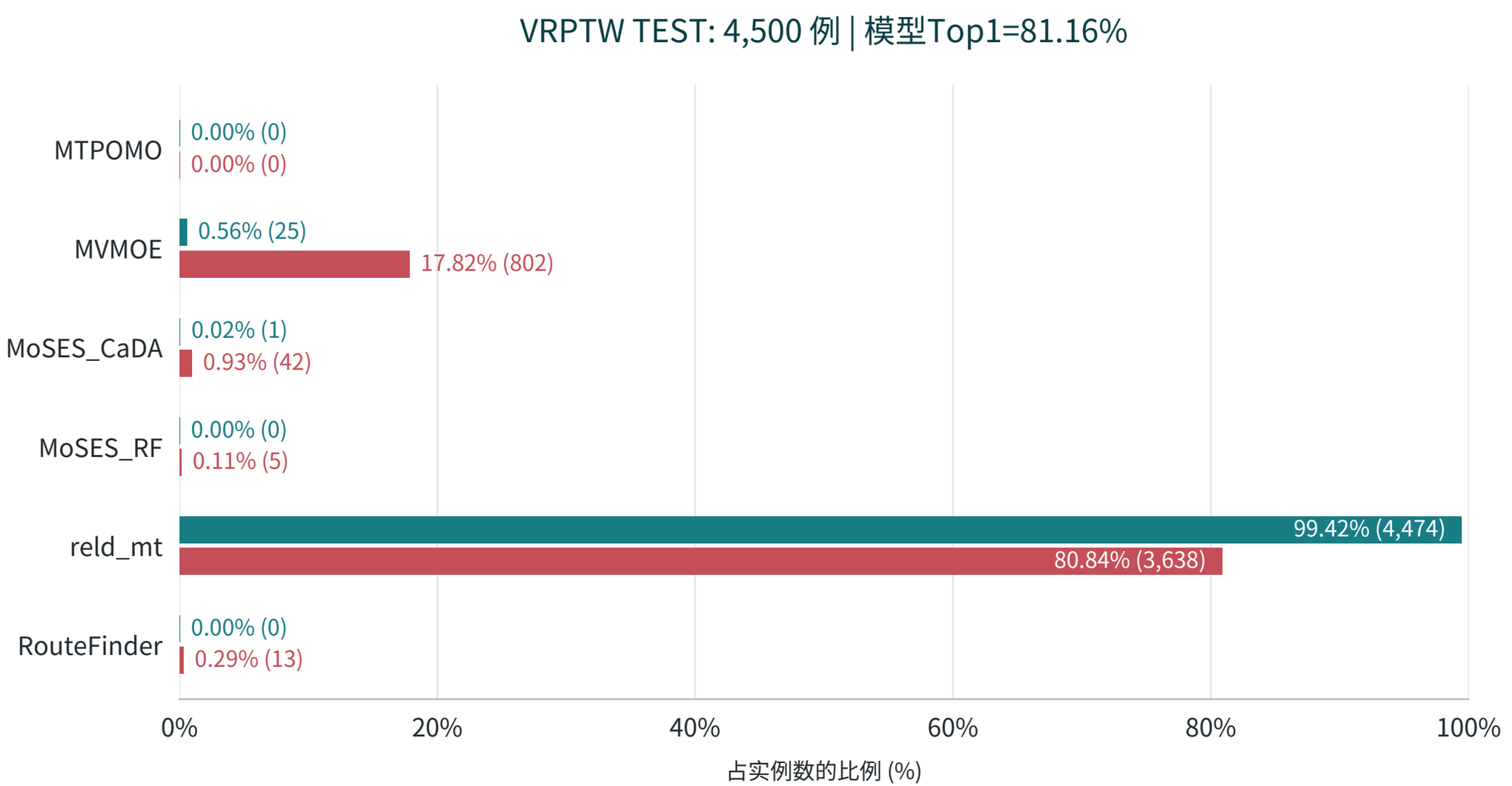
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

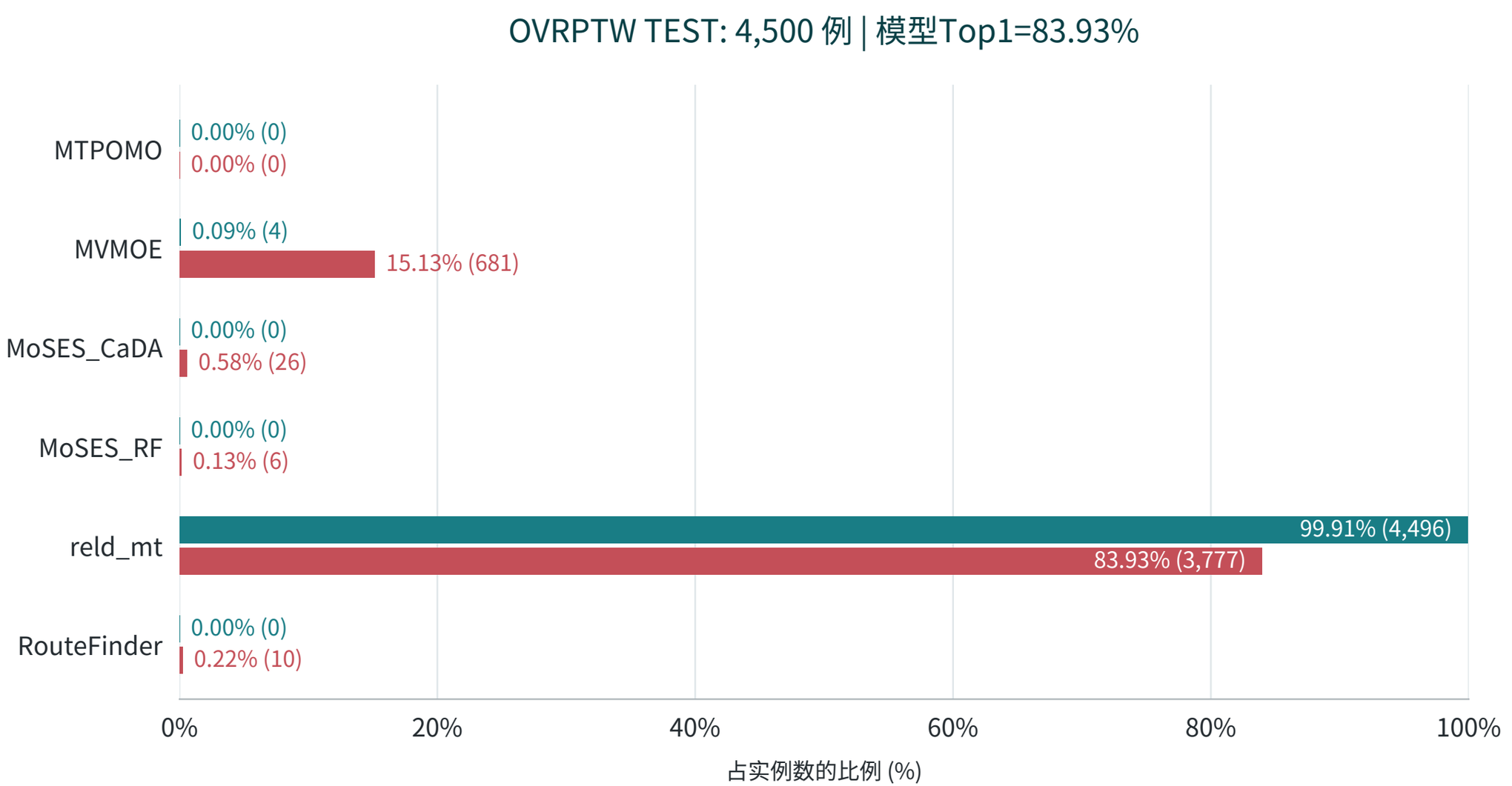
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

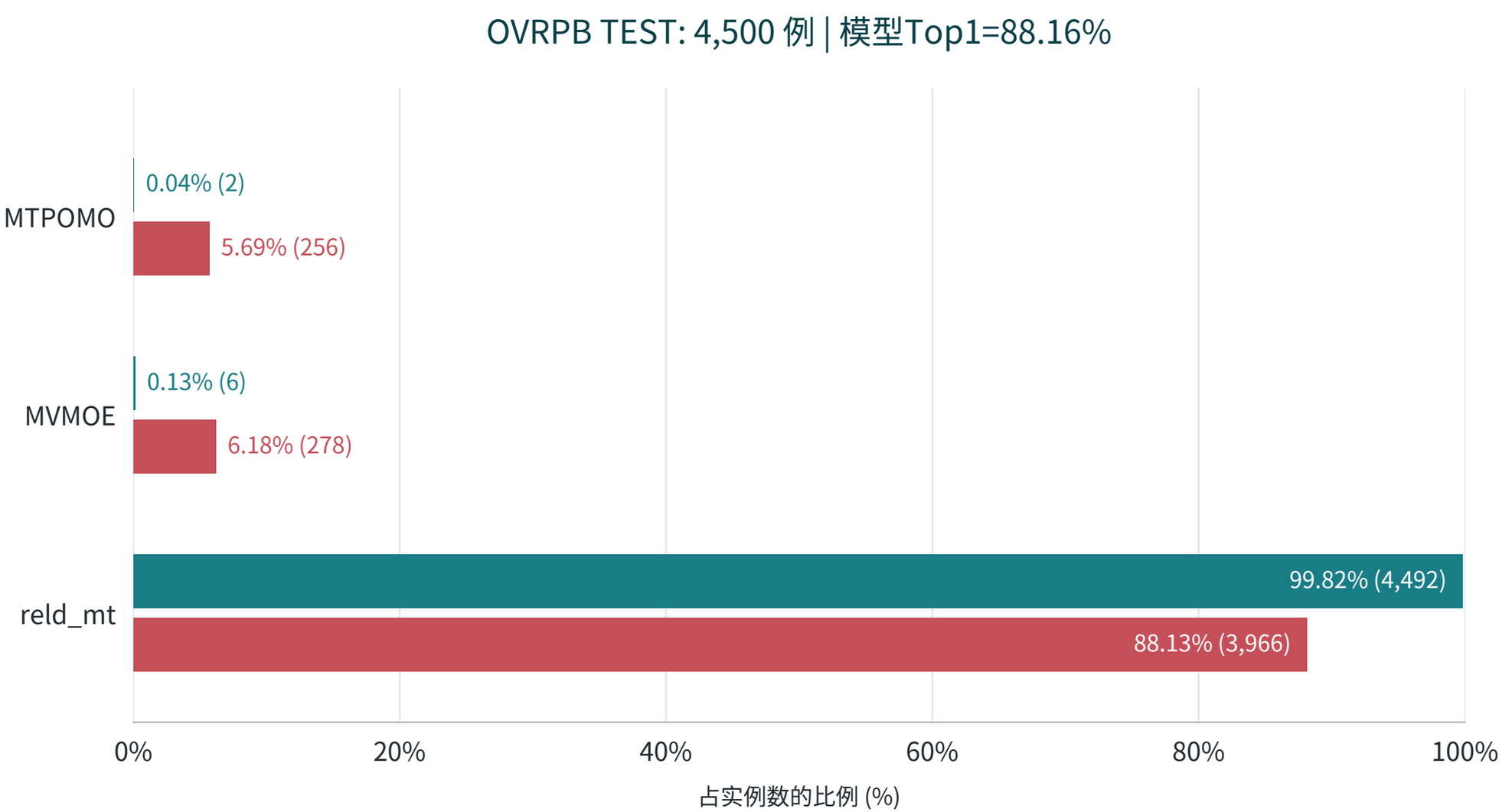
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPB | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

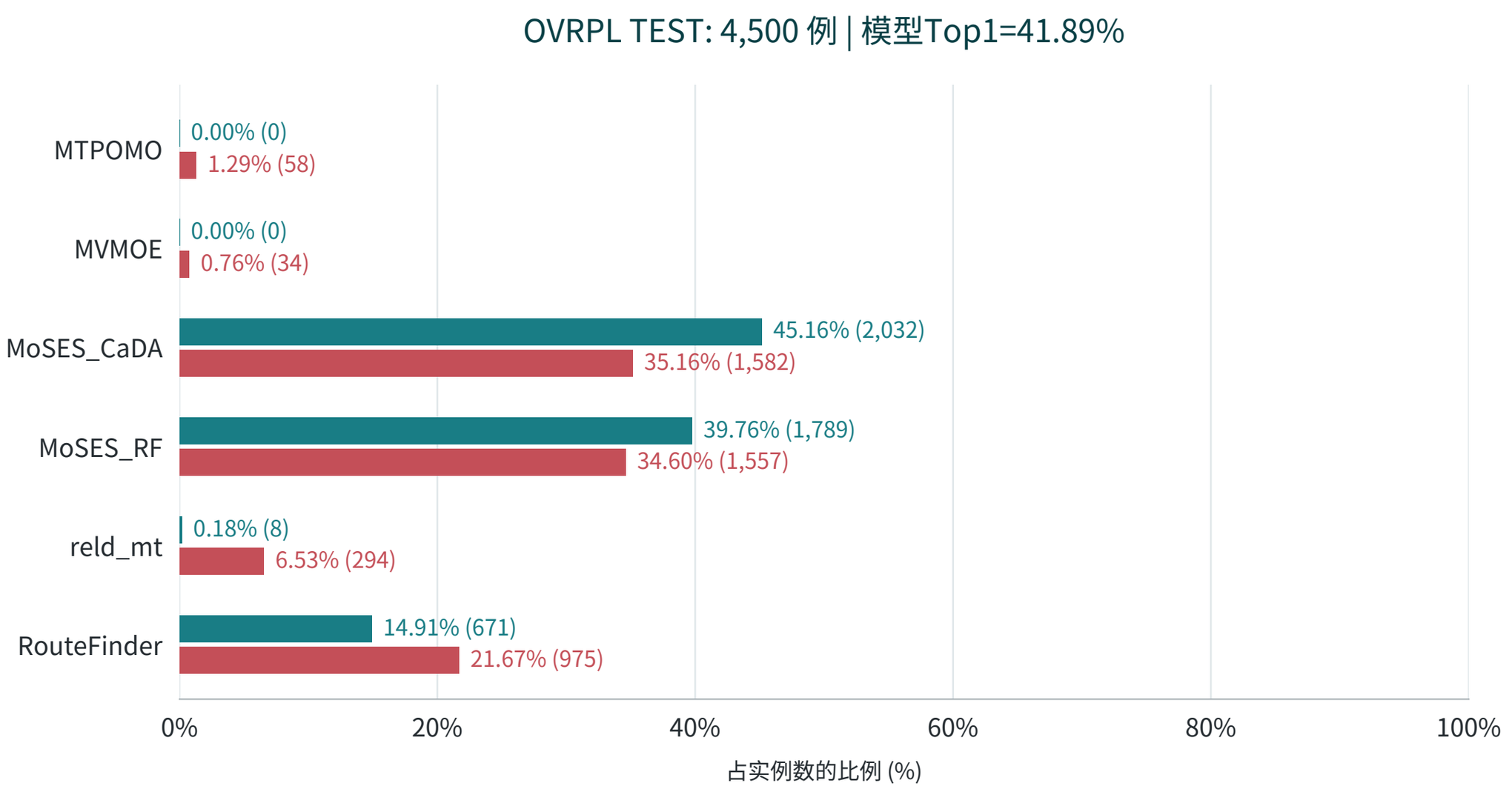
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | OVRPL | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

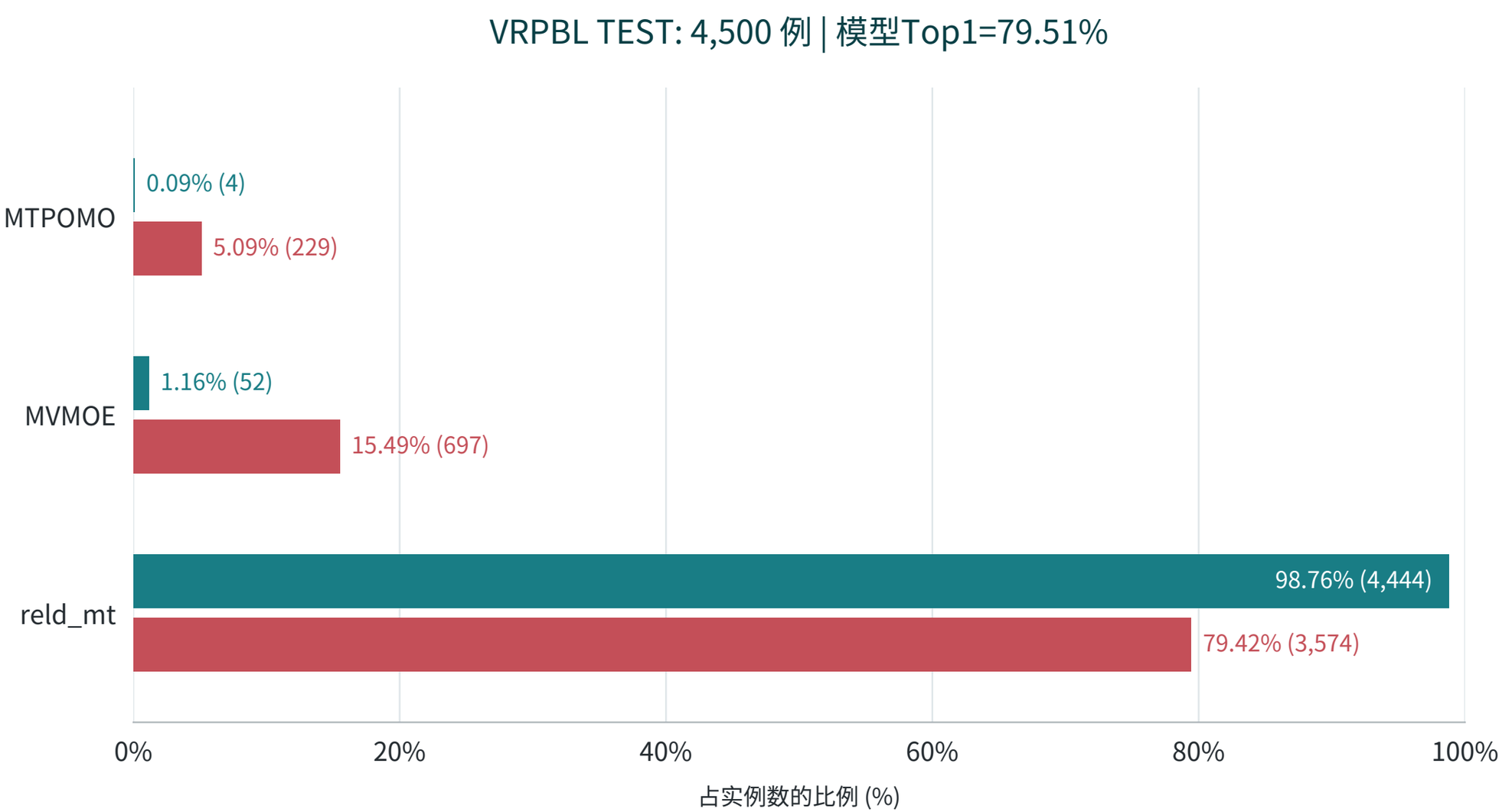
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPBL | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



标注: % (次数); 候选按 TEST 名称对齐，包含零次数候选。

模型Top1 = 100 * mean(pred == saved native winner); 不是最大选择频率。

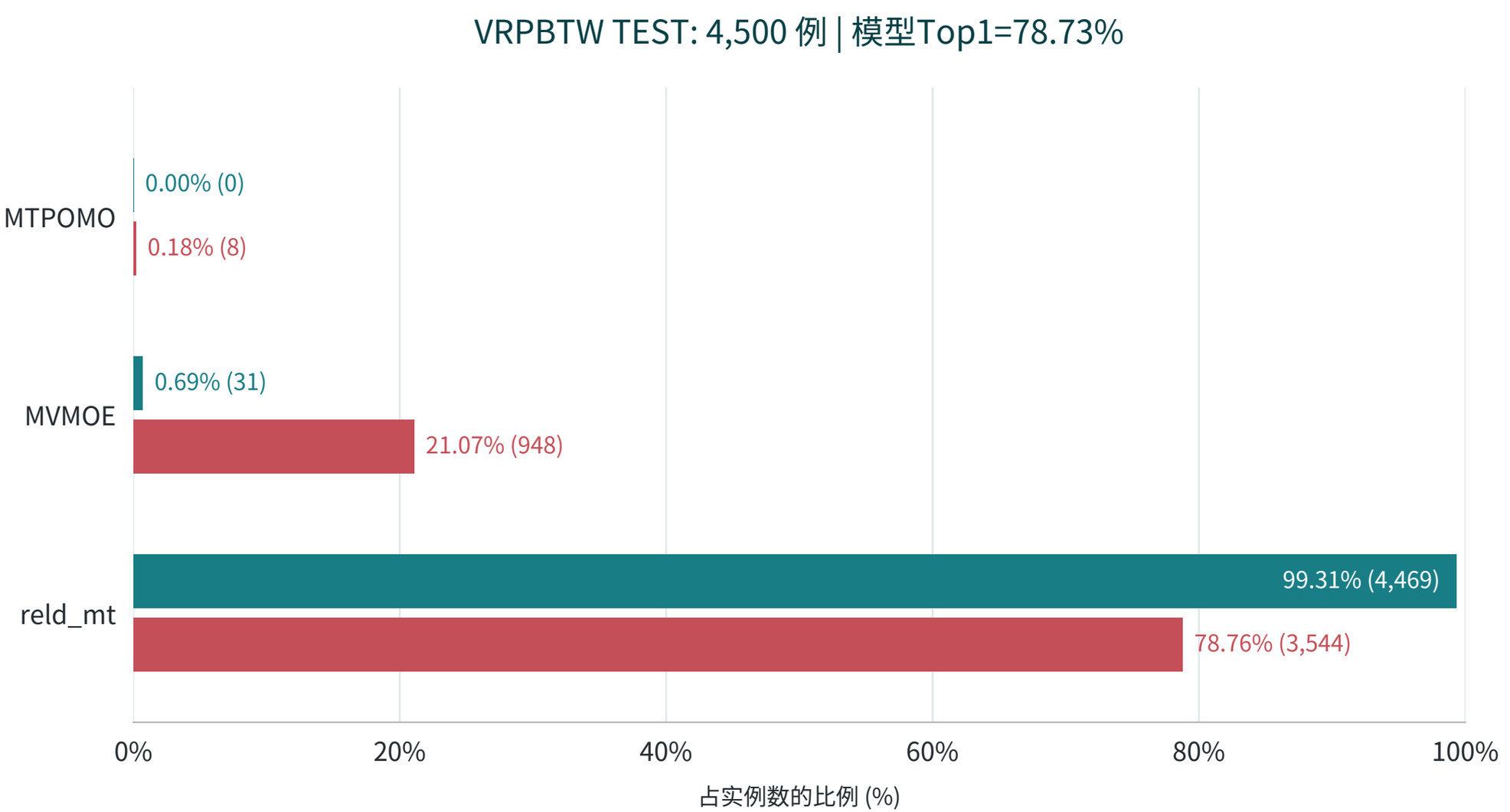
R101/A: stable global solver-ID score ties. Same R101 lock; LIB inference only; no training/reselection/Rxx mixing.

R101/A | VRPBTW | 方法分布

同一锁定模型: epoch 29 / update 41760 / SHA256 e96d42b96ec03155

新 TEST 数据版本: selection_data_new90k_B_filtered_single_reld_mt_seed2_20261018

标签: 原生 native/FP64 ind; 仅读锁定预测



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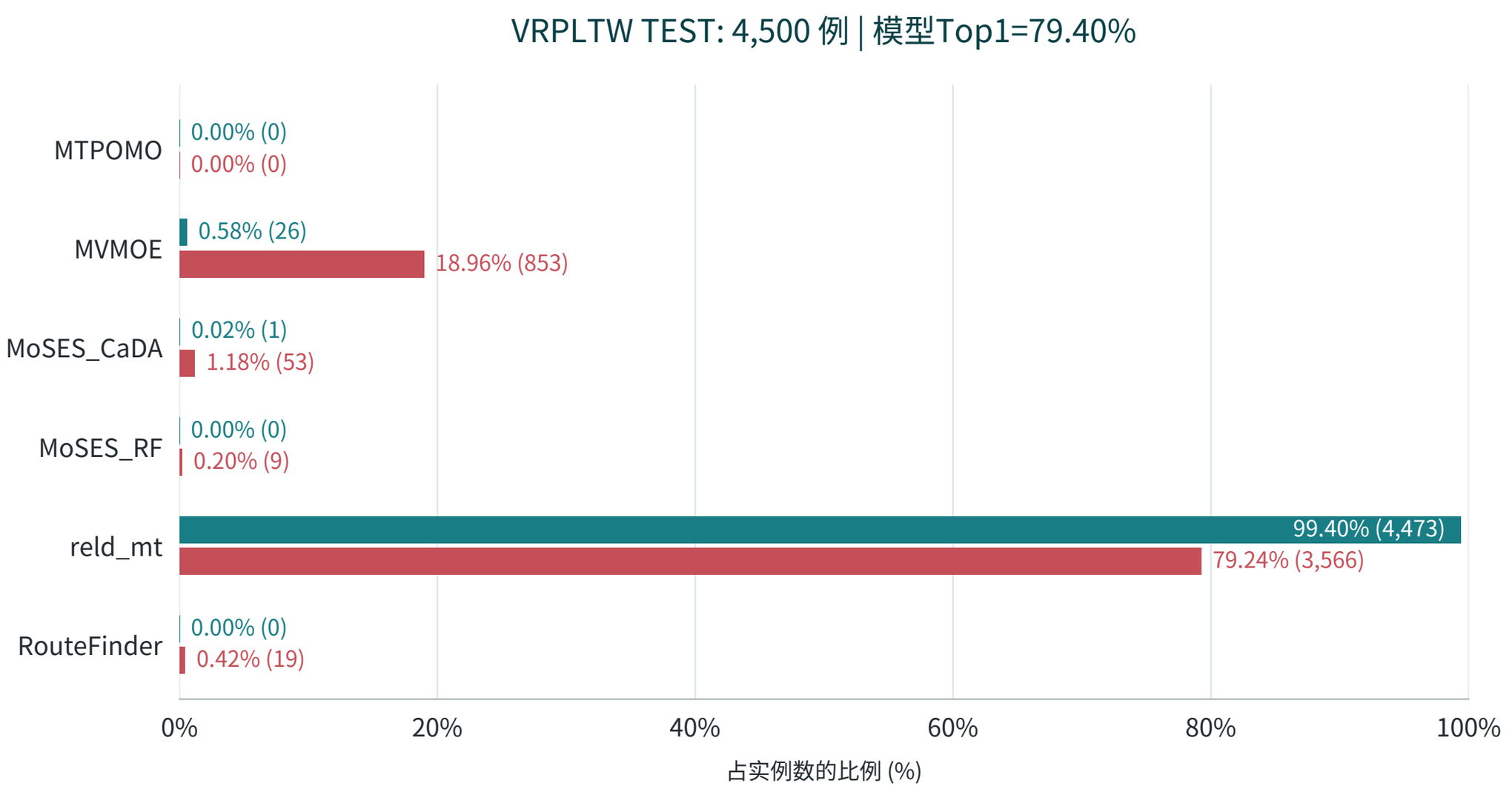
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R101/A | VRPLTW | 方法分布

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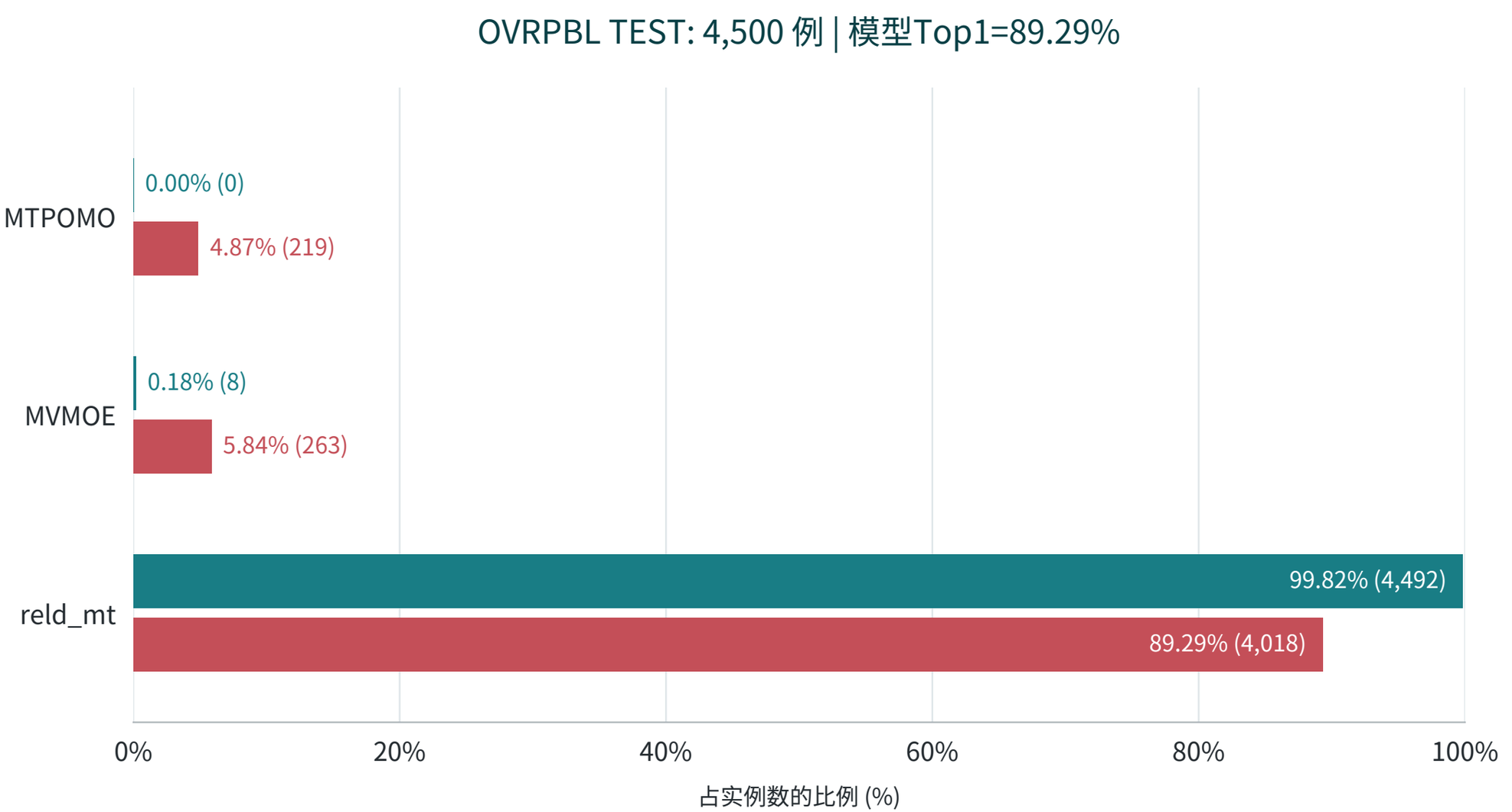
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R101/A | OVRPBL | 方法分布

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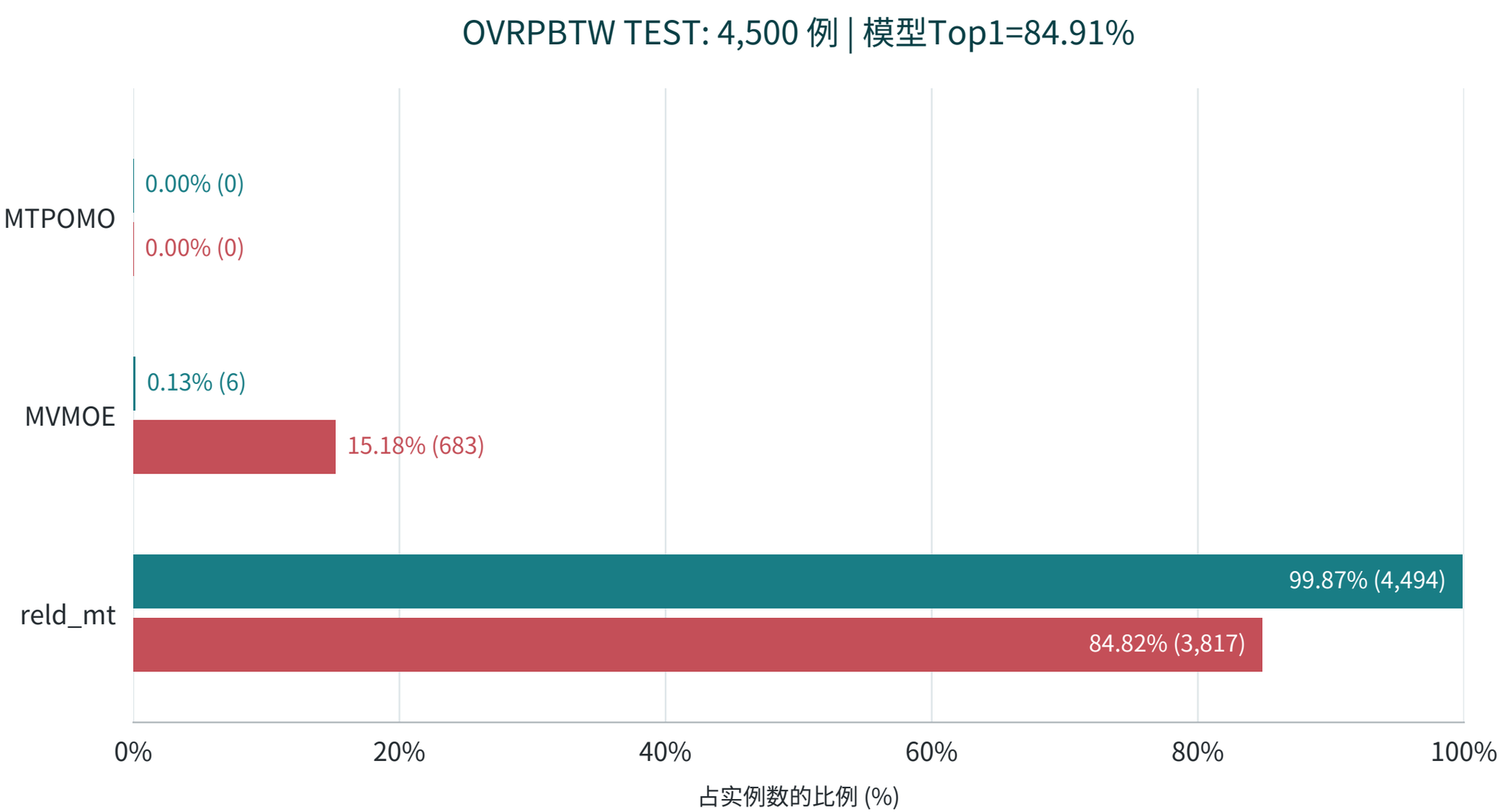
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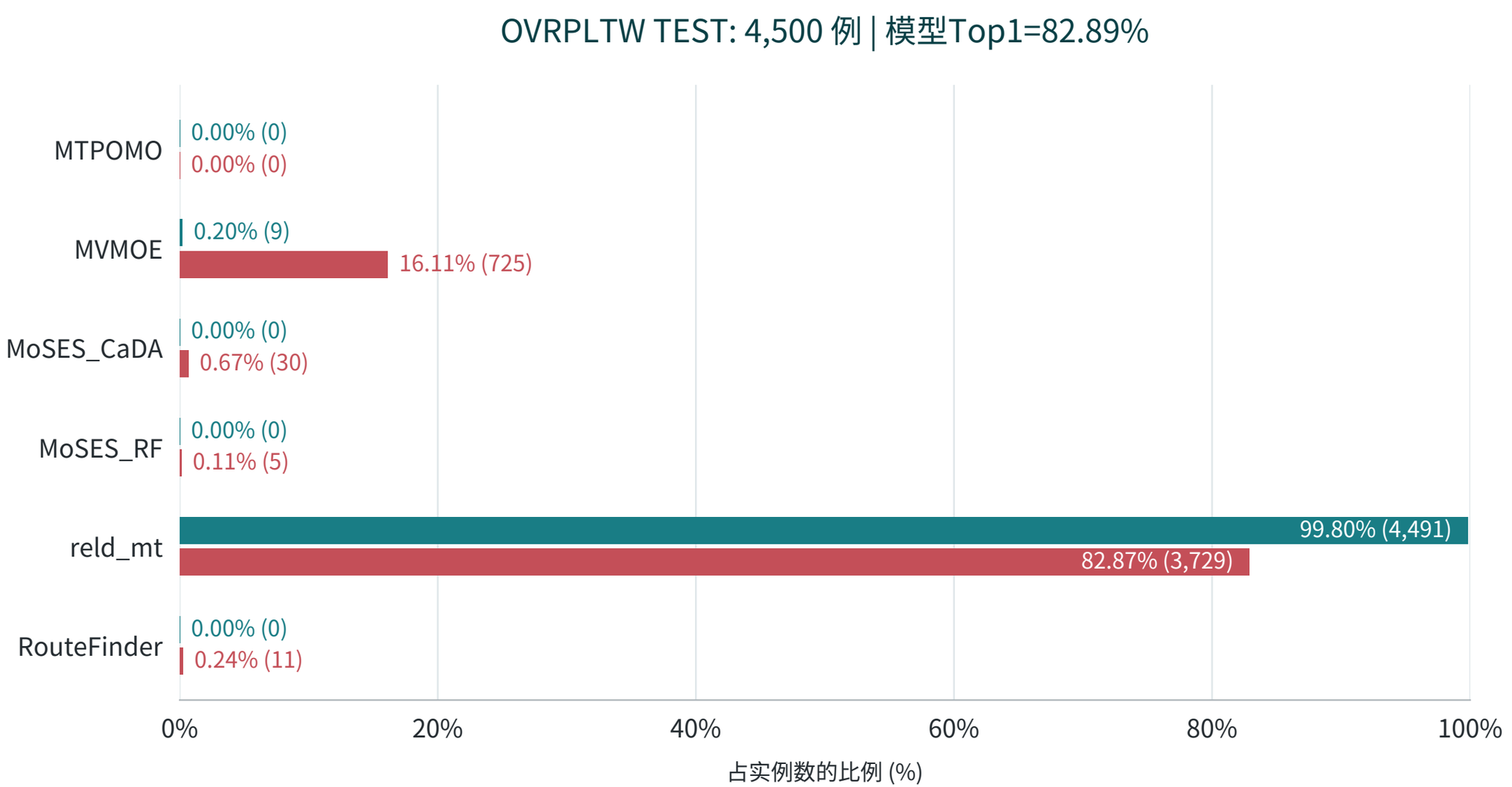
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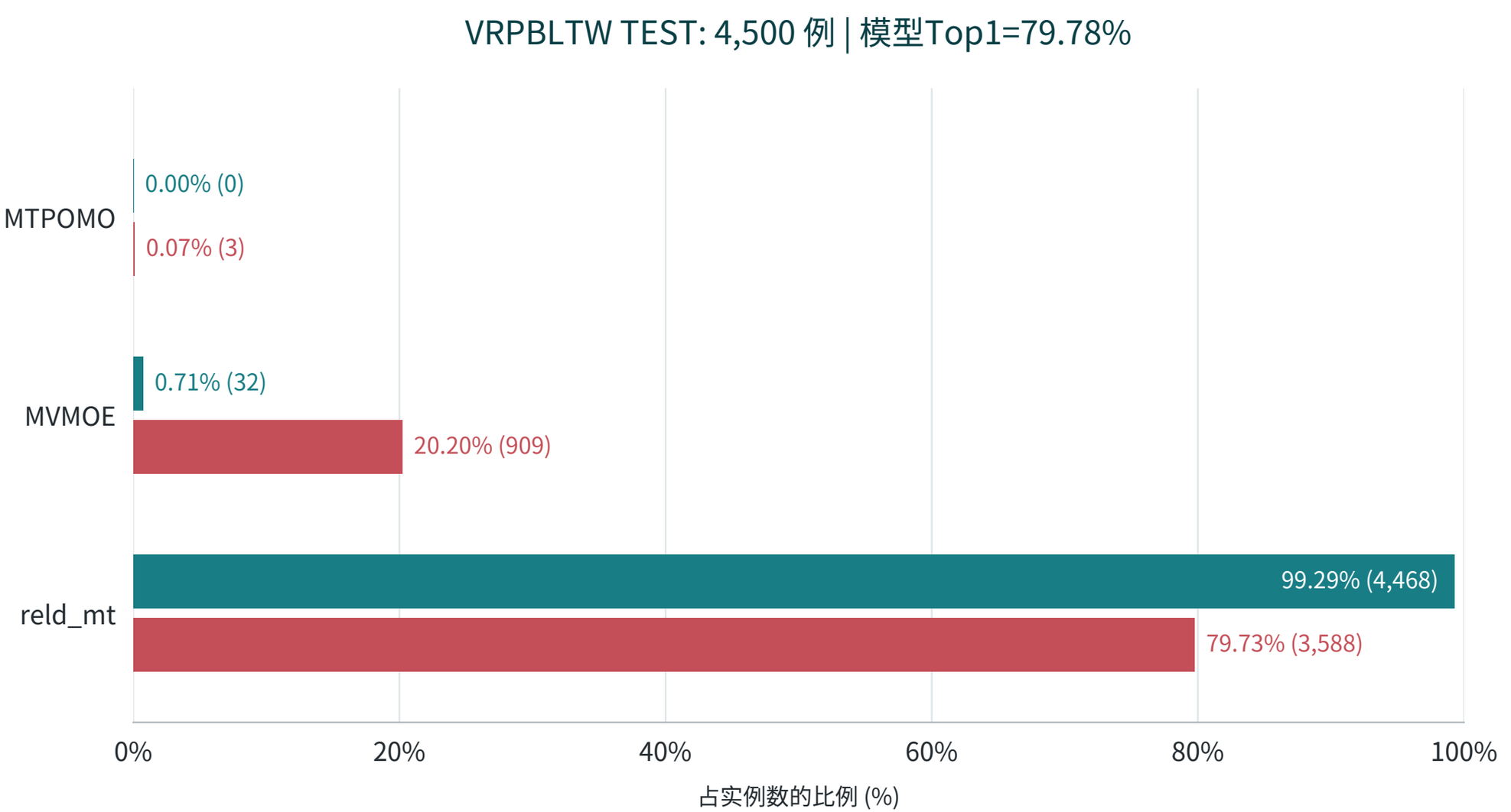
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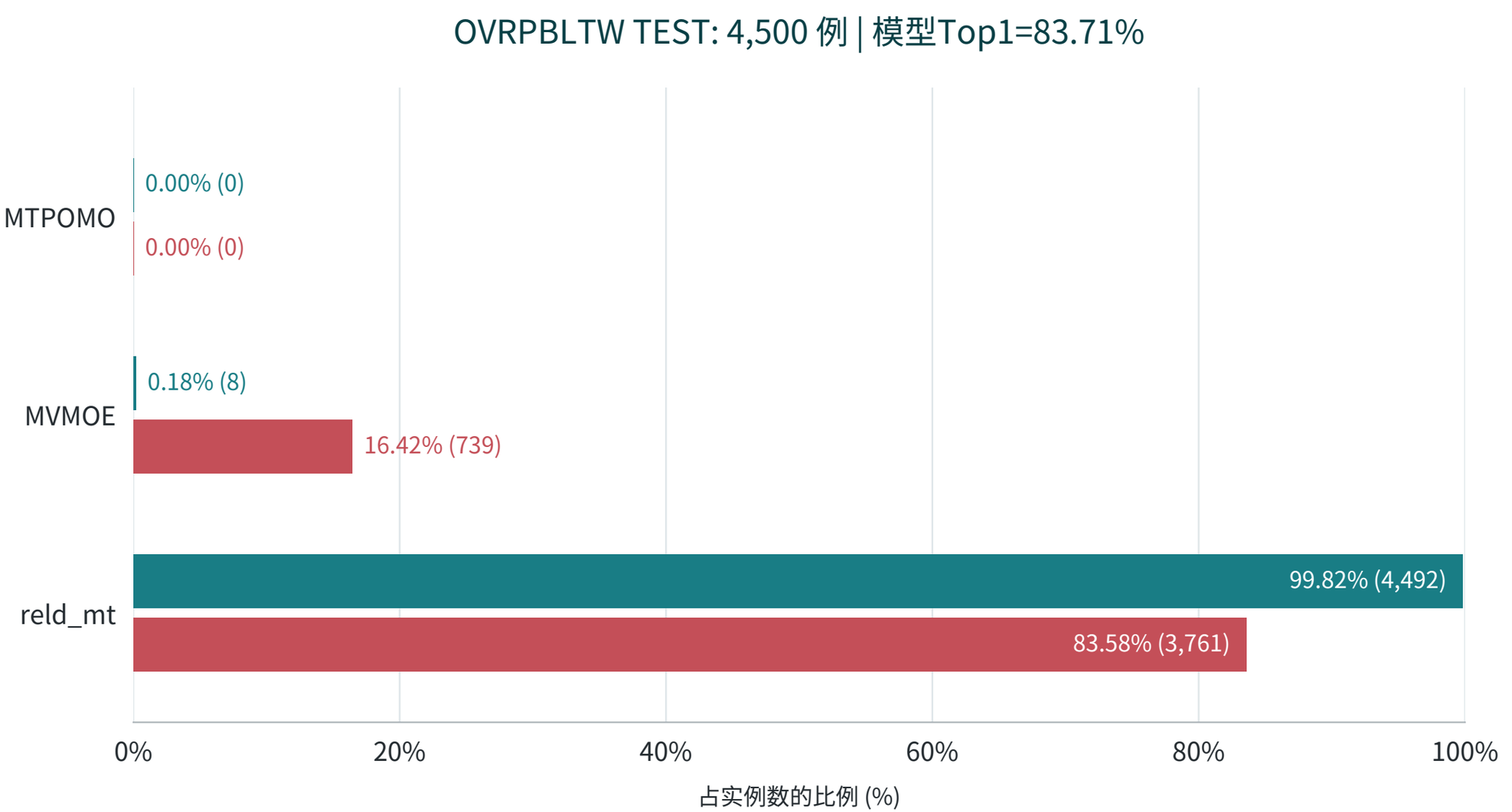
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